Bhushan Chhara Watershed

R-99

Figure 1:
Para (village) boundary
of Bhushan Chhara watershed

Disclaimer:
Boundaries are not necessarily authoritative

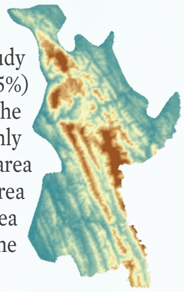
Source:
Center for Environmental and Geographic Information Services (CEGIS)

The Bhushan Chhara sub-watershed (R-99), located in the Bhushan Chhara Union, 158 no Mauza of the Barkal Upazila of Rangamati District, encompasses 20 paras (figure 1) and covers approximately 4,360 hectares of land. It is home to over 3,000 people, with communities consisting of the Chakma and Marma ethnic groups. The overall literacy rate is approximately 83%, yet opportunities for higher education are very limited. Farming is the main occupation. The average monthly income is 11,230 taka ($94), which is relatively low compared to the national average. About 99% of the houses are katcha and almost 93% of households use solar electricity systems. There is a noticeable scarcity of drinking water and a prevalent incidence of waterborne diseases.

Topography

Only 22% area lies below 50-meter (m) elevation (figure 2), which is situated in the north-east and south-west region of the watershed. Most of the study area lies between 50–80m (37%) and 80–120m (25%) elevation. The hilly ridge along the center of the watershed has comparatively higher elevation but only a little area falls under this class. About 11% of the area lies between 120–150m elevation and only 5% of the area lies between 150–200m elevation. Only 1% of the area lies above 200m elevation, which is situated in the central hilly region of the watershed.

Figure 2: Topography



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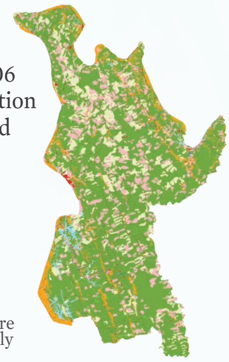
<50 • 100 • 150 • >200

meter Mean Sea Level (mMSL)

Landuse

Figure 3: Land use

A total 13 types of land use have been identified within the R-99 watershed (figure 3). The main land use is forest (2,806 hectares (ha)), followed by shifting cultivation (588 ha), herb/shrub (501 ha), and cropland (292 ha).



- Forest Tree Dominated Area (64%)
- Crop Land (7%)
- Herb/Shrub Dominated Area (12%)
- Rural Settlement (1%)
- Shifting Cultivation (13%)
- Orchard (1%)
- Others (2%)

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Demography

Population

50% 50%

Total 3,463

Literacy

Male 89% Female 77%

Average literacy rate 83%

Economy

11,230 Monthly Income (BDT)

11,253 Monthly Expense (BDT)

WASH

- 11 Gravity Feed
- 756 sanitary latrines
- 302 dug wells, 3 ring wells
- 60 chharas, 24 Springs (creeks & Springs)

Poverty

Very Poor	43%
Poor	35%
Middle Class	19%
Better Off	4%

10 50 100

Production

- Rice 3-5 metric tons per hectare (MT/acre)
- Fisheries 55-56 MT/acre
- 9,922 Livestock

Climatic Hazards

Climate Change-Induced Hazards

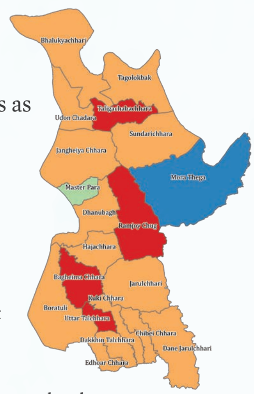
Nine climate hazards threaten Bhushan Chhara watershed agriculture, livelihood, ecosystem, and biodiversity. Drought, erratic rainfall, flash floods, heat waves, river floods, and lightning intensify across most of the paras. Landslides, cold waves, and hailstorms occur at low to moderate intensity.

Climate Change Risk

Figure 4: Mutihazard risk map

Climate change risk assessment identified 4 paras as very high-risk hotspots, 14 as high-risk, none as medium-risk, and the remaining paras as low to very low-risk (figure 4).

Very High	High	Medium	Low	Very Low
4	14	0	1	1



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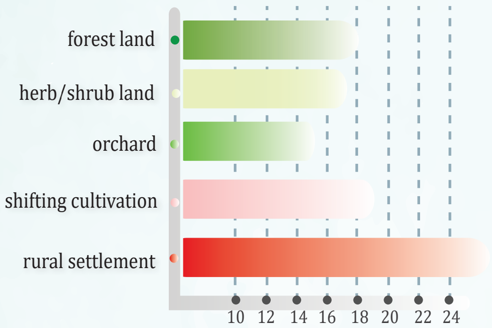
Watershed Degradation

Land Degradation Status

In R-99, land degradation is prominent in -



Causes of Degradation

- Jhum cultivation with short rotation
- Improper crop management
- Overgrazing
- Deforestation & overharvesting of trees
- Lack of vegetation cover
- Climate change
- Overuse of inorganic fertilizers

Degradation Triggering Factors

- Reduction of vegetation cover
- Quality and species composition change/diversity decline/monoculture plantation
- Soil erosion/landslides
- Surface sealing/hard pan development
- Waterlogging
- River bank erosion

Watershed Conservation

Conservation measures focused on enhancing ecological resilience as well as improving livelihood quality of the local community were identified and categorized into 6 classes.

Forest/ Shrub Land Conservation

20 ha Forest enrichment plantation

3 kilometer (km) Tree plantation along village road

Degraded Soil & Land Improvement

5 Landslide Treatment

15 Gully Treatment

Agriculture Land Conservation

40 ha Jhum to agroforestry

60 ha Fruit tree plantation

150 ha Orchard management

3 Drainage improvement

Stream Bank Protection

1 km Buffer strip plantation and bio-engineering methods

Water Resources Development

3 Irrigation canal improvement

5 Spring development

Capacity Development & Demonstration

40 Households (HHs) Regenerative farming training

20 HHs Drip irrigation training

20 HHs Homestead improvement

Watershed Management Infrastructures

Culvert Construction	1
Earthen Dam Construction	35
Dam Improvement	0
Irrigation Canal Construction	0
Protection Wall Construction	0

Watershed conservation interventions in Bhushan Chhara include dam construction (35) being the most implemented, aimed at enhancing water storage and distribution. In addition to these, other conservation measures have been identified to further boost water availability, agricultural productivity, and overall ecosystem health. Measures such as check dam construction, irrigation canal construction, culvert construction, and protection wall construction will help control runoff, reduce soil erosion, and protect critical infrastructure. Together, these interventions will significantly benefit the local community by ensuring year-round water access, improving crop yields, and reducing vulnerability to climate-induced hazards. These integrated efforts promote sustainable livelihoods and strengthen resilience in the Bhushan Chhara region.

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From the People of Japan
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